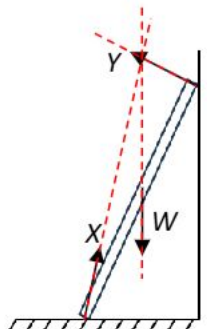3 **D** When the beam is in equilibrium, the lines of action of the 3 forces acting on the beam meet at a common point and there is no resultant moment about any point on the beam.